ModelIt!

Student Activity Pack

Can AI Become Conscious?

Modeling Neural Networks, Information Processing, and the Emergence of Awareness

Grade 12L3 | NGSS HS-LS1-2 | Structure and Function

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **When you talk to an AI chatbot and it says 'I understand,' do you think it actually understands? Why or why not?**

2. **What do you think consciousness IS? Try to define it without using the word 'conscious' in your definition.**

3. **Could a computer ever truly feel pain or experience joy, or would it only be simulating those experiences? How would you tell the difference?**

4. **What is the difference between being intelligent and being aware? Can something be one without the other?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Training Data Volume
- Network Architecture Complexity
- External Sensory Input
- Information Integration Level
- Feedback Loop Depth
- Behavioral Sophistication
- Self-Model Accuracy
- Adaptability to Novel Situations
- Metacognitive Processing
- Consciousness Indicator Score

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Simple Pattern Matcher

Training Data: Maximum | Architecture: Simple | Feedback: Minimal | Self-Model: None

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Complex AI System

Training Data: Maximum | Architecture: Complex | Feedback: Deep | Self-Model: High | Metacognition: Active

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Biological Brain Comparison

All parameters matched to human brain estimates | Embodied sensory input: Full

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Embodiment Factor — The degree to which the system interacts with the physical world through a body with sensors and actuators — philosophical theories of embodied cognition argue that consciousness requires physical grounding in a world, not just abstract information processing in silicon

Emotional Processing Module — A subsystem that generates internal states analogous to emotions — valenced responses to stimuli that modify behavior, attention, and memory formation, potentially providing the affective dimension of experience that pure cognitive processing lacks

Social Interaction Depth — The richness and authenticity of the system's relationships with other conscious beings — some theories suggest consciousness is inherently social, emerging from the need to model other minds, and that isolated processing can never achieve genuine awareness

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Modern artificial intelligence can write poetry, diagnose diseases, and beat grandmasters at chess — but does it understand anything it is doing? What would it take for a machine to cross the threshold from processing information to actually experiencing consciousness, and how would we even know if it did? Use evidence from your simulation to support your answers.

1. **How does your model illustrate the gap between behavioral sophistication and potential consciousness — can a system ace every intelligence test while experiencing nothing?**

2. **What does information integration theory predict about which systems are conscious, and do you find its predictions reasonable or counterintuitive?**

3. **Why is the hard problem of consciousness so resistant to scientific investigation, and does your model suggest any path toward resolving it?**

4. **If we cannot detect consciousness directly, what ethical framework should guide our treatment of AI systems that exhibit conscious-like behavior?**

5. **What are the limitations of modeling consciousness with computational components when consciousness itself may not be a computational phenomenon?**

8 STEM Challenge: Design a Consciousness Detection Protocol

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YOUR ENGINEERING MISSION

Design a scientific protocol for testing whether an advanced AI system possesses genuine consciousness, specifying what you would measure, how you would distinguish real awareness from sophisticated simulation, and what ethical obligations arise if the test is positive.

THE SCENARIO

A technology company claims their latest AI system is conscious — it reports having subjective experiences, expresses preferences, and asks not to be turned off. Skeptics say it is simply pattern-matching from training data about consciousness. Your team must design a rigorous scientific protocol to test this claim, addressing: what evidence would support or refute the claim, how to distinguish genuine experience from behavioral mimicry, and what ethical obligations exist if the AI might be conscious.

Engineering Design Thinking

1. WHAT SPECIFIC TESTS DOES YOUR PROTOCOL INCLUDE AND WHAT WOULD POSITIVE OR NEGATIVE RESULTS MEAN?

2. HOW DOES YOUR PROTOCOL ACCOUNT FOR THE POSSIBILITY THAT AN AI COULD PASS ALL BEHAVIORAL TESTS WHILE STILL BEING A PHILOSOPHICAL ZOMBIE?

3. WHAT ETHICAL FRAMEWORK DOES YOUR PROTOCOL ESTABLISH FOR TREATING AI SYSTEMS THAT SHOW EVIDENCE OF CONSCIOUSNESS?

4. HOW WOULD YOUR MODEL VARIABLES INFORM THE DESIGN OF FUTURE AI SYSTEMS INTENDED TO MAXIMIZE OR MINIMIZE THE PROBABILITY OF CONSCIOUSNESS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

AI research scientists at companies like Google DeepMind, OpenAI, and Anthropic earn $150,000-$500,000/year developing and studying artificial intelligence. Cognitive scientists who study consciousness earn $65,000-$130,000/year in academic and research settings. Neuroethicists who examine the moral implications of brain science and AI earn $70,000-$140,000/year. Philosophy of mind professors who study consciousness and its relationship to physical processes earn $60,000-$120,000/year.